PROBLEM 34

Particle on a Circular Wire in an Oscillating Vertical Plane

Original statement · Problems.pdf, p. 22

Oscillating Plane + Euler Force

A heavy point particle P of mass m slides without friction on a circular wire of center C and radius R. The wire is located on a vertical plane Oxy that rotates about the vertical axis Oy with a prescribed angular frequency $\boldsymbol{\Omega}(t) = A\omega\,\sin(\omega t)\,\mathbf{j}$, where A and ω are known positive constants, as shown in the figure. The angle that the vector $\overrightarrow{CP}$ makes with the Ox axis is denoted φ.

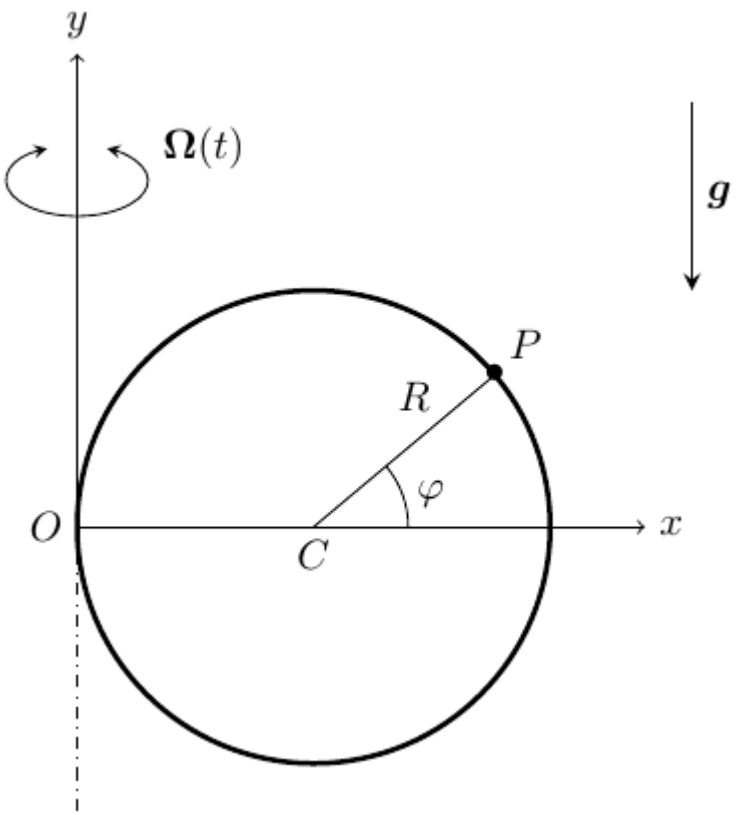


Figure: Circular wire of centre C and radius R in the vertical plane Oxy, passing through O, with C on the Ox axis; P on the wire at angle φ from Ox; the plane rotates about the vertical axis Oy with Ω(t); gravity g downward (Problems.pdf, p. 22).

Answer the following questions in a reasoned way:

(a) Write down the position, velocity and acceleration vectors of P relative to the reference frame $Oxyz$, as a function of φ and its derivatives.

(b) Identify the forces that act on P, and write down their mathematical expression for a generic instant of time t.

(c) Is the mechanical energy of P conserved as seen from the Oxy reference frame? Why?

(d) Write down a single second order differential equation to calculate φ as a function of time. Note that you do not need to solve this equation.

(e) Compute the reaction of the wire on the particle P as a function of φ, $\dot{\varphi}$ and t.

(f) Assume now that $A^2 \ll g/(R\omega^2)$ and that the motion of P is such that $\varphi = \delta - \pi/2$ with $\delta \ll 1$. Around what value of δ does the particle oscillate? Are there any values of ω for which the approximation $\delta \ll 1$ will not hold?

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** a heavy bead P (mass m) slides on a smooth circular wire of radius R whose centre C lies on the Ox axis at distance R from O (the circle passes through O). The vertical plane Oxy of the wire swings about the vertical axis Oy with the prescribed angular velocity $\mathbf{\Omega}(t) = A\omega\sin(\omega t)\,\mathbf{j}$.

- **Why the frame $Oxyz$.** The wire is fixed in $Oxyz$, so P is described by the single angle φ and its relative kinematics is that of circular motion. Because Ω depends on time, the inertia forces now include the **Euler force** $-m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P$ (Slide 28), which has no potential (Slide 31), and the centrifugal potential depends explicitly on time; the energy discussion of part (c) depends on both facts.
- **Hypotheses:**
  1. A frame S_0 with origin O on the fixed vertical axis is inertial; Oy is vertical upward and gravity is $-g\,\mathbf{j}$.
  2. The wire is smooth and bilateral: the reaction is perpendicular to the tangent of the wire, with a radial component in the plane of the wire and a component along $\mathbf{k}_1$, normal to the plane.
  3. The rotation of the plane is prescribed (imposed by an external drive whatever the motion of P).
- **Degrees of freedom:** one, the angle φ of $\overrightarrow{CP}$ measured from Ox.

Symbol	Meaning	SI unit
m	mass of P	kg
R	radius of the wire ($OC = R$)	m
A	amplitude parameter of $\Omega(t)$	1 (rad)
ω	frequency of the oscillation of the plane	rad/s
g	gravitational acceleration	m/s^2
φ	angle of $\overrightarrow{CP}$ from Ox	rad

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames.** $S_1 = Oxyz = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the plane of the wire ($\mathbf{i}_1$ along Ox, $\mathbf{j}_1$ along Oy upward, $\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ normal to the plane). $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial, with $\mathbf{j}_0 = \mathbf{j}_1$ (the rotation axis) and S_1 coinciding with S_0 at $t = 0$.
- **Angle turned by the plane.** With $\psi(t)$ the angle of rotation about $\mathbf{j}_0$, $\dot{\psi} = \Omega(t)$ and $\psi(0) = 0$. Integrating with the substitution $u = \omega\tau$, $du = \omega\, d\tau$, and Barrow's rule:

$$\psi(t) = \int_0^t A\omega \sin(\omega\tau)\, d\tau = A \int_0^{\omega t} \sin u\, du = A\,[-\cos u]_0^{\omega t} = A\,(1 - \cos \omega t)$$

The plane swings between $\psi = 0$ and $\psi = 2A$: it oscillates, it does not spin.

- **Change of basis** (simple rotation of angle ψ about $\mathbf{j}_0$, Notes Eqs. 1.18–1.20; columns are $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\psi & 0 & \sin\psi \\ 0 & 1 & 0 \\ -\sin\psi & 0 & \cos\psi \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\psi\cdot\cos\psi - 0 + \sin\psi\cdot\sin\psi = 1$$

The columns are unit vectors and pairwise orthogonal ($\cos\psi\sin\psi - \sin\psi\cos\psi = 0$), so $[{}_0R_1][{}_0R_1]^T = I$. Poisson check: $\left.\dfrac{d\mathbf{i}_1}{dt}\right|_0 = \dot\psi\,(-\sin\psi\,\mathbf{i}_0 - \cos\psi\,\mathbf{k}_0) = -\dot\psi\,\mathbf{k}_1$, and indeed $\Omega\,\mathbf{j}_1 \times \mathbf{i}_1 = -\Omega\,\mathbf{k}_1$.

- **Motion of S_1:** origin fixed on the axis, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$;

$$\boldsymbol{\omega}_{10} = \Omega(t)\,\mathbf{j}_1 = A\omega\sin(\omega t)\,\mathbf{j}_1, \qquad \boldsymbol{\alpha}_{10} = \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \dot\Omega\,\mathbf{j}_1 = A\omega^2\cos(\omega t)\,\mathbf{j}_1$$

($\mathbf{j}_1 = \mathbf{j}_0$ is fixed, and the chain rule gives $\frac{d}{dt}\sin\omega t = \omega\cos\omega t$.)

- **Polar basis of the wire** (fixed in shape in S_1): $\mathbf{e}_r = \cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1$ (along $\overrightarrow{CP}$), $\mathbf{e}_\varphi = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1$ (tangent), and $\mathbf{k}_1 = \mathbf{e}_r \times \mathbf{e}_\varphi$ ($\cos^2\varphi + \sin^2\varphi = 1$). Its matrix in $\mathcal{B}_1$ is a rotation about $\mathbf{k}_1$, with determinant $\cos^2\varphi + \sin^2\varphi = 1$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Relative position, velocity and acceleration

Pedagogical Justification: Relative quantities are derivatives in S_1, where $\mathbf{i}_1, \mathbf{j}_1$ are constant; only φ depends on time, so each component is differentiated with the chain rule.

$\mathbf{r}_1^P = \overrightarrow{OC} + \overrightarrow{CP}$ with $\overrightarrow{OC} = R\,\mathbf{i}_1$ and $\overrightarrow{CP} = R\,\mathbf{e}_r$:

$$\boxed{\mathbf{r}_1^P = R\,(1+\cos\varphi)\,\mathbf{i}_1 + R\sin\varphi\,\mathbf{j}_1}$$

With $\frac{d}{dt}\cos\varphi = -\sin\varphi\,\dot\varphi$ and $\frac{d}{dt}\sin\varphi = \cos\varphi\,\dot\varphi$:

$$\boxed{\mathbf{v}_1^P = -R\dot{\varphi}\sin\varphi\,\mathbf{i}_1 + R\dot{\varphi}\cos\varphi\,\mathbf{j}_1 = R\dot{\varphi}\,\mathbf{e}_\varphi}$$

Product rule on each component, e.g. $\frac{d}{dt}\left(-R\dot{\varphi}\sin\varphi\right) = -R\ddot{\varphi}\sin\varphi - R\dot{\varphi}^2\cos\varphi$:

$$\boxed{\mathbf{a}_1^P = -R\left(\ddot{\varphi}\sin\varphi + \dot{\varphi}^2\cos\varphi\right)\mathbf{i}_1 + R\left(\ddot{\varphi}\cos\varphi - \dot{\varphi}^2\sin\varphi\right)\mathbf{j}_1 = R\ddot{\varphi}\,\mathbf{e}_\varphi - R\dot{\varphi}^2\,\mathbf{e}_r}$$

(b) Forces acting on P

> **Pedagogical Justification:** In the non-inertial frame $Oxyz$ we use $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63). The actual forces are the weight and the reaction of the wire; the inertia forces follow from Slide 28 with $\mathbf{a}_0^O = \mathbf{0}, \boldsymbol{\omega}_{10} = \Omega\mathbf{j}_1, \boldsymbol{\alpha}_{10} = \dot{\Omega}\mathbf{j}_1$.

Actual forces:

- Weight: $\mathbf{W} = -mg\,\mathbf{j}_1$.
- Reaction of the smooth wire (no component along the tangent $\mathbf{e}_\varphi$): $\mathbf{N} = N_1\,\mathbf{e}_r + N_2\,\mathbf{k}_1 = N_1\cos\varphi\,\mathbf{i}_1 + N_1\sin\varphi\,\mathbf{j}_1 + N_2\,\mathbf{k}_1$.

Inertia forces (cross products with $\mathbf{j}_1\times\mathbf{i}_1 = -\mathbf{k}_1, \mathbf{j}_1\times\mathbf{j}_1 = \mathbf{0}, \mathbf{j}_1\times\mathbf{k}_1 = \mathbf{i}_1$):

- Origin: $\mathbf{F}_A = -m\mathbf{a}_0^O = \mathbf{0}$.
- Euler: $\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P = \dot{\Omega}\,\mathbf{j}_1\times\left[R(1+\cos\varphi)\mathbf{i}_1 + R\sin\varphi\,\mathbf{j}_1\right] = -\dot{\Omega}R(1+\cos\varphi)\,\mathbf{k}_1$, so

$$\mathbf{F}_E = -m\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P = mA\omega^2 R\cos(\omega t)\,(1+\cos\varphi)\,\mathbf{k}_1$$

- Centrifugal: $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = -\Omega R(1+\cos\varphi)\,\mathbf{k}_1$ and $\boldsymbol{\omega}_{10}\times\left(-\Omega R(1+\cos\varphi)\mathbf{k}_1\right) = -\Omega^2 R(1+\cos\varphi)\,\mathbf{i}_1$, so (Slide 30, distance to the axis $x = R(1+\cos\varphi)$)

$$\mathbf{F}_C = m\Omega^2 R\,(1+\cos\varphi)\,\mathbf{i}_1 = mA^2\omega^2 R\sin^2(\omega t)\,(1+\cos\varphi)\,\mathbf{i}_1$$

- Coriolis: $\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = \Omega R\dot{\varphi}\,\mathbf{j}_1\times\left(-\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1\right) = \Omega R\dot{\varphi}\sin\varphi\,\mathbf{k}_1$, so

$$\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = -2mA\omega R\dot{\varphi}\sin(\omega t)\sin\varphi\,\mathbf{k}_1$$

(c) Is the mechanical energy conserved in Oxy?

> **Pedagogical Justification:** By Slide 29, E_1^P is conserved only if every force that does work in S_1 derives from a potential that does not depend explicitly on time. Compute the power of each force in S_1.

- $\mathbf{N} \cdot \mathbf{v}_1^P = R\dot{\varphi}\,(N_1\mathbf{e}_r + N_2\mathbf{k}_1)\cdot\mathbf{e}_\varphi = 0$ (the wire is fixed in S_1).
- $\mathbf{F}_E$ and $\mathbf{F}_{COR}$ are along $\mathbf{k}_1$, normal to the plane in which $\mathbf{v}_1^P$ lies: they do no work. (In general the Euler force has no potential, Slide 31; here it happens to be powerless.)
- The weight has the potential $mgy = mgR\sin\varphi$.
- The centrifugal force has the potential $V_C = -\frac{1}{2}m\Omega(t)^2x^2$, but **it depends explicitly on time** because Ω does.

Define $E_1^P = \frac{1}{2}mR^2\dot{\varphi}^2 + mgR\sin\varphi - \frac{1}{2}m\Omega^2R^2(1+\cos\varphi)^2$. Differentiating with the chain and product rules,

$$\frac{dE_1^P}{dt} = mR^2\dot{\varphi}\ddot{\varphi} + mgR\cos\varphi\,\dot{\varphi} + m\Omega^2R^2(1+\cos\varphi)\sin\varphi\,\dot{\varphi} - m\Omega\dot{\Omega}R^2(1+\cos\varphi)^2$$

The first three terms are $mR^2\dot{\varphi}$ times the equation of motion of part (d), so they cancel, leaving

$$\frac{dE_1^P}{dt} = -m\Omega\dot{\Omega}\,R^2(1+\cos\varphi)^2 = -mA^2\omega^3R^2\sin(\omega t)\cos(\omega t)\,(1+\cos\varphi)^2 \neq 0$$

No: the mechanical energy of P in Oxy is not conserved, because the centrifugal potential changes with time (the frame does not rotate at a constant rate). It would be conserved if Ω were constant. Nor is the energy conserved in S_0: there the wire moves and its reaction $N_2\mathbf{k}_1$ does work.

(d) Equation of motion

> **Pedagogical Justification:** The tangential direction $\mathbf{e}_\varphi$ is the only one free of the unknown reaction ($\mathbf{N} \perp \mathbf{e}_\varphi$), and the Euler and Coriolis forces (along $\mathbf{k}_1$) also drop out.

Projections on $\mathbf{e}_\varphi = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1$: $m\mathbf{a}_1^P \cdot \mathbf{e}_\varphi = mR\ddot{\varphi}$; $\mathbf{W} \cdot \mathbf{e}_\varphi = -mg\cos\varphi$; $\mathbf{F}_C \cdot \mathbf{e}_\varphi = -m\Omega^2 R(1+\cos\varphi)\sin\varphi$. Hence $mR\ddot{\varphi} = -mg\cos\varphi - m\Omega^2 R(1+\cos\varphi)\sin\varphi$, i.e.

$$\boxed{\ddot{\varphi} + \frac{g}{R}\cos\varphi + A^2\omega^2\sin^2(\omega t)\,(1+\cos\varphi)\sin\varphi = 0}$$

a nonlinear equation with a time-periodic coefficient (parametric forcing). It agrees with the official key.

(e) Reaction of the wire

> **Pedagogical Justification:** The remaining two components of $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$, along $\mathbf{e}_r$ and $\mathbf{k}_1$, each contain one reaction component.

Along $\mathbf{e}_r$: $m\mathbf{a}_1^P \cdot \mathbf{e}_r = -mR\dot{\varphi}^2$; $\mathbf{W} \cdot \mathbf{e}_r = -mg\sin\varphi$; $\mathbf{F}_C \cdot \mathbf{e}_r = m\Omega^2 R(1+\cos\varphi)\cos\varphi$. So $-mR\dot{\varphi}^2 = N_1 - mg\sin\varphi + m\Omega^2 R(1+\cos\varphi)\cos\varphi$:

$$\boxed{N_1 = mg\sin\varphi - mA^2R\omega^2\,(1+\cos\varphi)\sin^2(\omega t)\cos\varphi - mR\dot{\varphi}^2}$$

Along $\mathbf{k}_1$: $m\mathbf{a}_1^P \cdot \mathbf{k}_1 = 0 = N_2 + mA\omega^2 R\cos(\omega t)(1+\cos\varphi) - 2mA\omega R\dot{\varphi}\sin(\omega t)\sin\varphi$:

$$\boxed{N_2 = 2mAR\omega\,\dot{\varphi}\sin\varphi\sin(\omega t) - mAR\omega^2\,(1+\cos\varphi)\cos(\omega t)}$$

and $\mathbf{N} = N_1\cos\varphi\,\mathbf{i}_1 + N_1\sin\varphi\,\mathbf{j}_1 + N_2\,\mathbf{k}_1$. Both components agree with the official key. The normal component N_2 is what the wire must supply to drag P along with the swinging plane (it balances the Euler and Coriolis forces).

(f) Small oscillations about the lowest point

> **Pedagogical Justification:** For $\delta \ll 1$ the trigonometric functions are replaced by their first-order Taylor expansions; the condition $A^2 \ll g/(R\omega^2)$, i.e. $A^2\omega^2 \ll g/R$, lets us drop the parametric term $A^2\omega^2 \sin^2(\omega t)\,\delta$ against $(g/R)\,\delta$. The result is a linear forced oscillator.

With $\varphi = \delta - \pi/2$: $\cos\varphi = \cos(\delta - \pi/2) = \sin\delta \approx \delta$, $\sin\varphi = \sin(\delta - \pi/2) = -\cos\delta \approx -1$, $\ddot{\varphi} = \ddot{\delta}$. Substituting into the equation of part (d) and keeping first-order terms:

$$\ddot{\delta} + \frac{g}{R}\,\delta - A^2\omega^2 \sin^2(\omega t)\,(1 + \delta) = 0$$

Dropping $A^2\omega^2 \sin^2(\omega t)\,\delta$ (small compared with $(g/R)\,\delta$) and using $\sin^2(\omega t) = \frac{1}{2}\,[1 - \cos(2\omega t)]$:

$$\boxed{\ddot{\delta} + \frac{g}{R}\,\delta = \frac{A^2\omega^2}{2}\,[1 - \cos(2\omega t)]}$$

Solution. With $\omega_n = \sqrt{g/R}$, a particular solution is a constant plus a term at 2ω. The constant δ_0 satisfies $\omega_n^2 \delta_0 = A^2\omega^2/2$. For $\delta_p = C\cos(2\omega t)$, $\ddot{\delta}_p = -4\omega^2 C\cos(2\omega t)$ and $(\omega_n^2 - 4\omega^2)\,C = -A^2\omega^2/2$. Hence

$$\delta(t) = \frac{A^2 R\omega^2}{2g} - \frac{A^2\omega^2}{2\,(\omega_n^2 - 4\omega^2)}\cos(2\omega t) + a\cos\omega_n t + b\sin\omega_n t$$

(a, b set by the initial conditions). The particle oscillates, with natural frequency $\omega_n = \sqrt{g/R}$, about

$$\boxed{\delta_0 = \frac{A^2 R\omega^2}{2g}}$$

which is small, consistent with $A^2 \ll g/(R\omega^2)$. It is positive: on average the bead is displaced slightly outward, away from the axis, by the mean centrifugal force.

Breakdown. The forced amplitude $|C| = \dfrac{A^2\omega^2}{2\,|\omega_n^2 - 4\omega^2|}$ grows without bound when $2\omega \to \omega_n$, i.e. for

$$\boxed{\omega \approx \frac{1}{2}\sqrt{\frac{g}{R}}}$$

At exact resonance the particular solution is $\delta_p = -\dfrac{A^2\omega^2}{4\omega_n}\, t\sin(\omega_n t)$ (check: $\ddot{\delta}_p + \omega_n^2\delta_p = -\frac{A^2\omega^2}{2}\cos\omega_n t$), whose amplitude grows linearly in time. The assumption $\delta \ll 1$ then fails, as stated in the key.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Role of each inertia force.** The centrifugal force (along $\mathbf{i}_1$, in the plane) is the only inertia force that enters the equation of motion. The Euler and Coriolis forces are normal to the plane: they do not change φ, but they load the wire through N_2.
- **Limits.** $A \to 0$: $\ddot{\varphi} + (g/R)\cos\varphi = 0$, the simple pendulum about $\varphi = -\pi/2$ in a fixed plane, and $N_2 = 0$. $\omega \to 0$ with A fixed: $\Omega \to 0$ and the same pendulum is recovered.
- **Energy.** dE_1^P/dt averages to zero over a period of the forcing ($\sin\omega t\cos\omega t$ has zero mean) but is not zero instant by instant; near resonance the drive pumps energy into the oscillation.
- **Inertial check.** Differentiating $[{}_0R_1]\mathbf{r}_1^P$ twice reproduces $\mathbf{a}_1^P + \boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P + \boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P) + 2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$ (Notes Eq. 3.27), and $m\mathbf{a}_0^P = \mathbf{W} + \mathbf{N}$ holds with the reaction of part (e) (checked symbolically with sympy).
- **Dimensions.** $[g/R] = [A^2\omega^2] = \mathrm{s}^{-2}$ (same units as $\ddot{\varphi}$); $[mA^2R\omega^2] = [mR\dot{\varphi}^2] = [mg] = \mathrm{N}$; $[mAR\omega\dot{\varphi}] = \mathrm{kg\cdot m\cdot s^{-2}} = \mathrm{N}$; $[A^2R\omega^2/g] = \dfrac{\mathrm{m\cdot s^{-2}}}{\mathrm{m\cdot s^{-2}}} = 1$, as an angle must be.

Source: Problems.pdf (PDF page 22); key on PDF page 75